

Product and Weather Data Load

Requirement:

This case study is part of Databricks L2 training. This case study covers the following steps:

- a. Reading Product data csv file
- b. Creating and loading Product data as a delta table
- c. Incremental data load for product delta table
- d. Streaming data load of weather data using delta table

Technology Used: Databricks Spark Platform (Python notebooks)

Assignment 1:

Create a delta table for storing Product data and perform incremental insert / update

Input data files: product.csv, product_delta.csv

(Double click on each data file in DataFiles.doc to open it and save it to your system locally)

Steps Involved:

Step1: Read Product.csv as a dataframe and save it as a delta table

Step2: Read Product_delta.csv as a dataframe and perform an upsert transaction on the product delta table

Step3: Verify that new records are inserted and existing records are updated in the delta table

Assignment 2:

Load the data from multiple weather stations into Spark delta lake tables.

Dataset: Weather data files (WS01.txt, WS02.txt WS10.csv)

(Double click on each data file in DataFiles.doc to open it and save it to your system locally)

Steps Involved:

Step1: Create a folder and copy the weather history files from above link to a folder in databricks file system.

Step2: Read the weather data files one at a time using streaming data read, ensuring one file is read per trigger. (Hint: use readstream with maxFilesPerTrigger = 1)

Step3: Perform the following transformations

- a. Drop the following columns : record_min_temp_year, record_max_temp_year
- b. Filter out records with actual_precipitation = 0
- c. Create a new column deviation_from_avg = abs(actual_precipitation – avg_precipitation)

Step4: Write the data to a weather delta table which is partitioned by year and month

Assignment 3:

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(This assignment is Optional as it requires azure subscription)

Modify Assignment 1 to store data files on to azure blob storage and access it from databricks VM.

Step 1: Create a storage account and container and upload the files

Step 2: Create Shared Access Signatures with read and write permissions

Step3: Create a Key Vault and secrets for sas token

Step4: Create a Databricks Workspace in Azure

Step5: Modify the code in assignment 1 to access the files from Azure blob storage

Documents to be shared as part of solution:

- a. Source code
- b. Sample output for each assignment